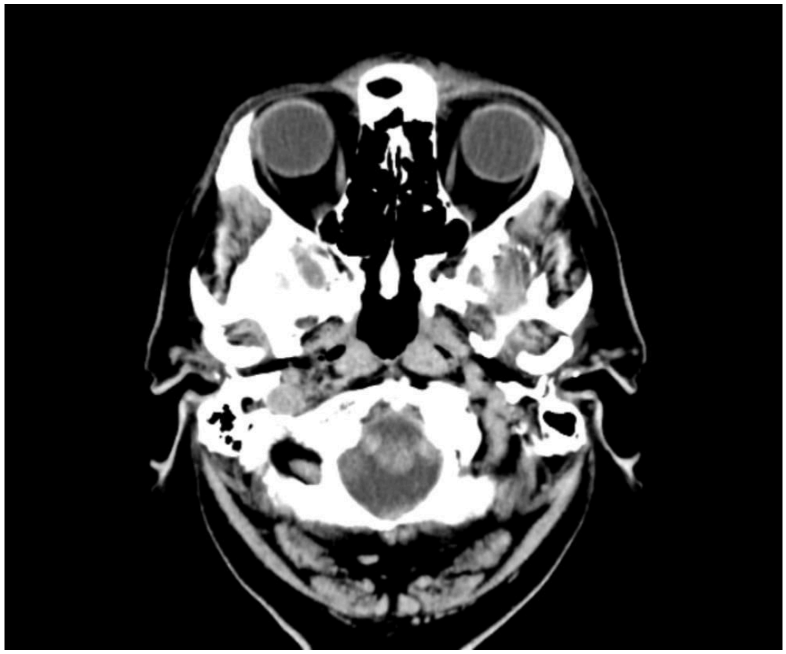
Lab 6

(Show input and output side by side for all problems)



1. **Compress** the above images using Discrete Cosine Transform (DCT), Haar Transform, and DCT-Haar, and find out which one is better in terms of compression ratio and PSNR for the given images.

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2. Apply Gaussian noise to Figure 1, and then use the following to restore the image:

   i. Geometric Mean filter
   ii. Harmonic Mean filter
   iii. Contra-harmonic Mean filter

3. Apply Gaussian noise to Figure 1, and then use the following order statistic filters to restore the image:

   i. Median filter
   ii. Maximum filter
   iii. Minimum filter
   iv. Midpoint filter
   v. Alpha-trimmed filter
   vi. Trimmed filter

4. By observing and comparing each of the outputs, determine which filter restores the image closest to its original state. Mention the reasoning behind your observation.

Figure 1: Tumor -1